# BASANT BHATTARAI

**DATA & AI ARCHITECT / SENIOR DATA ENGINEER**

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## PROFILE

Data and AI architect and senior data engineer with 9+ years on production systems in industry. Sets the target architecture, chooses platforms on measured evidence, designs the data model and the governance and SOC 2 controls around it, then leads the build. Builds on open-source engines — Kafka, Spark, Iceberg, Trino, dbt and Airflow on Kubernetes — and the databases underneath: modeling, replication, backup and recovery, upgrades. AI built into the platform: agent workflows on LangGraph, Google ADK and MCP whose output lands as typed, reviewable rows, with isolation, evaluation and review designed in. Track record: 40% lower infrastructure cost, 50% faster processing, 60% faster analytics delivery.

## EXPERIENCE

### Senior Data Engineer | UXCam

Feb 2024 - present

- Own the architecture, reliability and governance of a platform handling 15 TB+ and 100M+ events daily: freshness and latency objectives per dataset, 99.9% uptime, incident-ready runbooks, reviewable data contracts, a named owner for every published table, and the SOC 2 criteria met for the platform and its infrastructure through every audit.
- Design and operate the production agent workflows — LangGraph and Google ADK orchestration, MCP tool servers, retrieval and evaluation — built into the existing platform, so model output lands as typed, provenanced rows rather than a transcript: analytics delivery 60% faster with 75% less analyst effort, review and fallback paths kept.
- Operate the SQL and NoSQL databases underneath: replication, backup and recovery, access, tuning, upgrades.
- Lead architecture and schema reviews before new stores, pipelines and agent workflows reach production.
- Mentor 5 engineers through pairing, code review and written design feedback; team delivery time down 30%.

### Data Engineer | UXCam

Feb 2020 - Feb 2024

- Rebuilt the core ETL layer in Spark and PySpark with Airflow orchestration, cutting processing time for 5 TB+/day by 50% and unblocking real-time product dashboards.
- Made change and recovery safe by treating schema evolution, replay, late-arriving data and backfills as designed interfaces with idempotent loads and explicit watermarks, not as emergency procedures.
- Tuned Trino, ClickHouse, Citus and TimescaleDB serving 100M+ queries/day for a 50% p95 latency reduction through partitioning, clustering and model redesign.
- Led the Kubernetes migration of the data infrastructure: 40% lower infrastructure cost, 99.9% uptime, and autoscaling that held under peak load.

### Project Leader | SVCET, India

2019 - 2020

- Led 4 developers building an intelligent dialogue system on an NLP and deep-learning stack, reaching the client's acceptance threshold for intent recognition and delivering 2 weeks ahead of schedule.
- Coordinated requirements, planning and review across academic and industry stakeholders, and introduced the code-review and version-control discipline the department kept afterwards.

### Software Developer | SV Technology, India

2017 - 2019

- Built Python and FastAPI backend services carrying 10K+ users at 85% test coverage.
- Designed and maintained PostgreSQL schemas, stored procedures and reporting paths, and set up CI with Jenkins alongside early pipeline work, the start of the move into data engineering.

## SELECTED OPEN SOURCE

- **data-platform-architecture** and **ai-platform-architecture** (reference architectures): a SaaS company leaving Postgres, measured: 18 decision records, a Kimball model, SOC 2 readiness, a cost model and a Databricks proof of concept; then its usage assistant on governed data.
- **spark-sql-kinesis** (maintainer): Kinesis source and sink for Spark 4.2 on AWS SDK v2, on Maven Central; an idle shard costs one GetRecords call, not a multi-second floor per batch.
- **evalproof**: LLM evaluation with judge calibration and CI gates that fail only on drops that are both large and significant.
- **distributed-db-lab**: Citus sharding, online rebalance and tenant isolation, measured against TimescaleDB and replicated ClickHouse. Also data-stack-lab, a governed agent, streaming, real-time ML, retrieval, data contracts and GitOps delivery. github.com/technobasant

## CAPABILITY MAP

### Data and AI architecture

Platforms chosen on evidence, with every decision written down and AI built in.

Target architectures and platform selection, conceptual to physical modelling, Kimball and Type 2 history, lakehouse, ClickHouse and Citus layouts, contracts in CI, decision records, SOC 2 and erasure by design, tenant isolation outside the model, semantic layers, release gates, cost models.

### Data platforms

Consumer-facing reliability, governance, recovery and operational readiness.

Python, SQL and Scala; Kafka and Kinesis ingestion (maintainer of an open-source Spark 4 Kinesis connector); Spark/PySpark processing, including the Spark 4 migration; Airflow orchestration; Iceberg, Hive Metastore, dbt and lakehouse modeling; Databricks with Unity Catalog; Kubernetes, Docker and Terraform delivery; contracts, lineage, cost and recovery.

### AI systems

Model output that lands as typed, reviewable rows rather than a transcript.

LangGraph, LangChain, Google ADK, MCP servers, Pydantic, Milvus and MLflow; bounded tools, retrieval, evaluation before rollout, provenance, human review and honest fallback paths.

### Database lifecycle

Modeling, replication, failover, backup and PITR, upgrades and query performance.

PostgreSQL, ClickHouse, Trino, Citus, TimescaleDB, Redis and MongoDB; partitioning, replication, restore and high availability.

### Technical leadership

Design review, mentoring, runbooks, and making system ownership transferable.

Architecture and operational review, mentoring, runbooks, service and freshness objectives, Prometheus and Grafana.

### Cloud

**AWS:** S3, EKS (IRSA), Lambda, Glue, Athena, Redshift, Kinesis, EMR | **GCP:** BigQuery, Dataflow, Dataproc, Datastream, Pub/Sub, Cloud Composer, GKE, Vertex AI | **Azure:** Synapse, Data Factory, Databricks, ADLS, Event Hubs, AKS, Azure ML

## EDUCATION AND LANGUAGES

### Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

English: C1 | Nepali: native

**Selected work, writing and case studies: basantbhattarai.com.np/work/**